22EGMO4

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Problem

Given a positive integer $n \geq 2$, determine the largest positive integer N for which there exist $N+1$ real numbers $a_0, a_1, \ldots, a_N$ such that

(1) $a_0 + a_1 = -\frac{1}{n}$, and

(2) $(a_k + a_{k-1})(a_k + a_{k+1}) = a_{k-1} - a_{k+1}$ for $1 \leq k \leq N-1$.

Solution. We claim that $N \leq n$.

Let $b_k = a_{k-1} + a_k$. Replacing this, we get:

(i) $b_1 = \frac{-1}{n}$, and

(ii) $b_k b_{k+1} = b_k - b_{k+1}$ for $1 \leq k \leq N-1$.

The second condition transforms to $b_{k+1} = \frac{b_k}{b_k + 1}$.

We can recursively conclude that $b_i = \frac{-1}{n-(i-1)}$. As a result, b_{n+1} is undefined.

So, $N \leq n$.

For the construction of $N = n$, consider the following sequence $\{a_k\}$:

(i) $a_0 = 0$

(ii) $a_{i+1} = \frac{-1}{n-i} - a_i$ for $0 \leq i \leq n-1$. $\square$